

## A FRÖHLICH INTERACTION OF HUMAN ERYTHROCYTES

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Experiments on human red cells provide evidence for an interaction compatible with the quantum mechanical theory of Fröhlich.

Fröhlich has a theory [1,2] that long-range coherent interactions, known to exist in the solid state, should occur in living cells in a unique form. Necessary conditions of the theory [2] are (a) a cell membrane potential difference, (b) a supply of metabolic energy, and (c) organisation, a type of order which, according to Fröhlich "... exists in a well-working machine in contrast to a broken-down one [2]". In work on human red cell rouleau formation [3–6] we noticed a small attraction between erythrocytes in plasma prior to the actual contact which precedes doublet formation. When we (a) reduce the membrane potential near to zero or (b) deplete the cells of ATP or (c) derange the cellular organisation with glutaraldehyde, the attraction can no longer be detected.

Membrane potentials act across very short distances ( $\approx 10$  nm) which implies that the membrane is strongly polarised electrically. Vibrations of membrane molecules will be dipolar, representing an electric wave of half wavelength equal to the thickness of the membrane [2]. If the velocity of mechanical membrane waves is assumed to be  $10^3$  ms $^{-1}$ , the frequency of such oscillations will be of the order of  $10^{11}$  Hz. There is evidence for such oscillations in microorganisms [2].

Fröhlich considers two basic collective excitations. The first, quasi-static, is of metastable states of very high electric polarisation. He then introduces metabolic energy which can excite coherent vibrations. If the supply of energy to the modes of a band of electric polarisation waves exceeds a minimum value, the energy may be channelled into a single mode, giving rise to

a strong coherent oscillation extending throughout the cell. Collective vibrations are quantised (phonons) and obey Bose–Einstein statistics. Fröhlich likens the excitation of a single mode to the Einstein condensation of a Bose gas [7], a condensation in momentum space, not in coordinate space as in liquefaction. In the quasi-static case interactions between highly polarised cell constituents will be repulsive and of short range. Fröhlich posits quite a different situation at  $\approx 10^{11}$  Hz when ionic screening is only significant at distances of micrometres. Moreover interactions may be attractive or repulsive as conditions change. Cooper [8] has applied the theory to the movement of macromolecules in the complex processes of cell mitosis.

In our experiments, heparinised blood at a haematocrit of  $\approx 0.5\%$  in a haemocytometer chamber is photographed every 15 s with a time-lapse 16 mm ciné-microscope for 3 h. About 200 erythrocytes are in view. In an early frame of the developed film, pairs of cells less than a cell diameter apart ( $< 8 \mu\text{m}$ ) are identified and their subsequent motion is followed until a doublet forms or until the end of the film. Isolated cells execute brownian movement. If the motion of pairs is also random, simple geometry predicts a maximum probability of the pair making contact of 0.5 if the initial separation is zero, to 0.15 at an initial separation of  $8 \mu\text{m}$ . The average probability is about 0.3. It will be seen from table 1 that in six experiments on four subjects and involving 212 pairs, the fraction of pairs which collided was  $0.85 \pm 0.06$ .

The pH of the plasma was reduced with 0.14 M HCl

Table 1  
Fraction of pairs of cells initially  $< 8 \mu\text{m}$  apart which collide within 3 h.

| Subject | Pairs     | Collisions | Fraction                   |
|---------|-----------|------------|----------------------------|
| SR      | 15        | 14         | 0.93                       |
| SR      | 56        | 46         | 0.82                       |
| SR      | 21        | 18         | 0.86                       |
| DLF     | 63        | 47         | 0.75                       |
| LSS     | 23        | 20         | 0.87                       |
| REL     | 34        | 30         | 0.88                       |
|         | 212 total | 175 total  | 0.85 mean (sd $\pm 0.06$ ) |

in 0.14 M NaCl and restored to normal with 0.14 M NaOH. At pH = 6.3 the membrane potential is close to zero [9,10]. Pair collision is reduced to the random value but is restored to a non-random value on returning the same cells to pH = 7.6 (table 2). Ionophore A23187 induces rapid entry of  $\text{Mg}^{++}$  and depolarisation of membrane potential [11]. Cells were washed in standard phosphate-buffered saline (PBS) containing  $10 \mu\text{M}$  A23187 and then suspended in plasma to which had been added 10 mM  $\text{MgCl}_2$ ,  $10 \mu\text{M}$  EGTA and  $10 \mu\text{M}$  A23187 [11]. Pair collision is reduced to just within the random range by this procedure (table 2). Cells were metabolically depleted by washing in PBS followed by incubation in PBS for 22 h at  $37^\circ\text{C}$  [12,13]. Such cells also give a result in the random range and the same cells are restored nearly to the mean fraction for normal cells by incubation for a further 3 h in 30 mM adenosine (table 2). Glutaraldehyde crosslinks protein and so "disorganises" the cell [14]. Cells were washed twice in PBS containing 0.04% glutaraldehyde, resuspended in the same solution and shaken gently for 24 h

at  $22.0^\circ\text{C}$ . The mean fraction is well within the theoretical random range (table 2) but it is likely that in these embalmed cells the membrane potential and metabolic mechanisms were abolished. Latex microspheres suspended in plasma were used as controls (table 2).

Brownian motion is monitored in each experiment by measuring the  $X$  coordinate against time of ten or more cells which remain isolated ( $> 8 \mu\text{m}$ ) throughout 3 h. From theory [15]

$$\langle X^2 \rangle = 2kTt\mu^{-1},$$

where  $\mu$  is the drag coefficient. Correlation coefficients averaged  $0.94 \pm 0.03$  (for microspheres  $r = 0.96$ ) but  $\mu$  (including that for microspheres) was consistently higher than the value calculated from Stoke's law or from the rate of free fall of erythrocytes [16]. So cells, normal and experimentally modified, and artificial particles when resting on a surface, experience a drag greater than that of the suspending medium.

By simple experiments we have demonstrated a hitherto unknown interaction of human erythrocytes and tested it against the Fröhlich theory. The most satisfactory of our tests is the reversible disappearance of the interaction by abolition of the membrane potential. Metabolic depletion does more than cut off the supply of energy. Although reversible it modifies structure [17] and results in ion shifts [18]. The zeta potential, however, is unchanged [12]. We measured intra- and extra-cellular chloride before and after metabolic depletion and, from the Nernst equation [10], calculated the membrane potential. It dropped from 15.0 mV to 7.1 mV and so might have fallen beneath some threshold for the Fröhlich effect. Glutaraldehyde

Table 2  
Fraction of pairs of erythrocytes initially  $< 8 \mu\text{m}$  apart ( $< 3.4 \mu\text{m}$  for microspheres) which collide within 3 h.

|                                      | No. of experiments | Mean fraction |
|--------------------------------------|--------------------|---------------|
| { cells at pH 6.3                    | 3                  | 0.33          |
| { cells restored to pH 7.6           | 3                  | 0.76          |
| ionophore-treated cells              | 2                  | 0.48          |
| { metabolically depleted cells       | 2                  | 0.39          |
| { cells repleted with adenosine      | 1                  | 0.75          |
| glutaraldehyde fixed cells           | 5                  | 0.37          |
| $3.4 \mu\text{m}$ latex microspheres | 1                  | 0.24          |

has a rather drastic effect on structure although some enzyme activity remains [14]. The theory regards organisation, membrane potential and energy supply as separate but the three entities are likely to be functionally interwoven and may prove difficult to isolate. Fröhlich forces will be universal in living systems. We offer our preliminary findings to enable other workers to check and extend them.

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